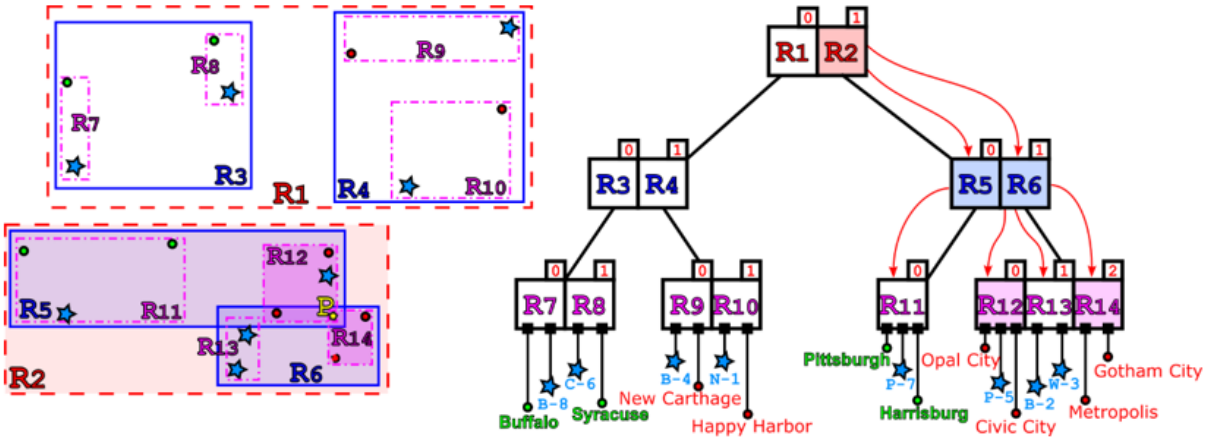


Figure 10.8 Unsuccessful search on the R-tree in figures 10.4-10.5. The path of the search is highlighted in both the Cartesian and tree views, and red arrows show the branches traversed in the tree. Notice the compact representation of the tree, with respect to figure 10.5.

This means we need to go through the children of both rectangles R5 and R6, checking from R11 to R14. Of these, only R12 and R14 can contain P, so those are the only rectangles whose points we will check, at the last step: neither contains P, so the search method can return false, and optionally the two leaves' rectangles that could host P, if inserted.

Nearest neighbor search works similarly, only instead of checking if a point belongs to each rectangle, it keeps the distance of the current nearest neighbor, and checks if each rectangle is closer than that (otherwise, it can prune it). This is to all extents similar to the rectangular region search in k-d trees, as described in section 9.3.6.